TRIS S. BROOKE-CUNNINGHAM

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PROFESSIONAL EXPERIENCE

Ally Financial – Auto & Insurance Product Delivery **February 2024 – Present**
Full Stack Web Developer *Detroit, MI*

- Collaborated with product owners to create detailed design documents and Jira acceptance criteria, enabling the development of complex, transaction-based application UIs.
- Developed dealership-facing UIs for *SmartCash*, a platform handling millions of dollars weekly by facilitating trades, loans, and financing for dealership inventory, using tools like Storybook, Jira, and Figma.
- Prototyped, demoed, and refined UIs for entering and viewing transactions, managing dealer and branch profiles, and overseeing vehicle inventories, ensuring alignment with Ally's UX team design specifications and *SmartCash's* business requirements.
- Modernized *SmartCash's* front-end, transitioning from Struts II and JSP to a ReactJS front-end and a Java based back-end, for improved performance security, and maintainability, and ease of use.
- Performed accessibility testing for *SmartCash* with Axe DevTools, Lighthouse, and NVDA, ensuring compliance with accessibility standards and improving user experience for all users.
- Automated unit testing, security scans, code quality checks, and deployments to dev, CIT, and UAT environments using GitLab pipelines.
- Increased development velocity by using proprietary GPT and LLama based AI copilot models to perform code generation reduce duplication of code, catch inefficient practices, and generate comprehensive unit tests for react UI components, rest controllers, and batch processes.

University of Michigan - Digital Studies Institute **August 2022 – January 2024**
Technical Coordinator & Programming Assistant *Ann Arbor, MI*

- Created technical documentation computers, consoles, VR technology, 360° conference equipment, and multimedia displays in the office and organized those resources for ease of deployment, allowing immediate use for faculty and staff.
- Communicated technical needs between faculty, teaching staff and administration to ensure tech purchases could be made that were effective for staff while still being within the set budget.
- Trained new staff on event planning & technical documentation procedures, setting up a framework for position continuity.

PROJECTS

CertPathway | *Capstone Project*

- Designed the code management system for Certpathway's codebase, used gitflow to minimize merge conflicts, and created interfaces for the different modules that made up the project to ensure clean integration at key milestones.
- Set deadlines for the team and used Trello for team management to successfully meet technical goals ahead of deadlines while managing proportional workloads without overlapping with team members' other constraints.
- Implemented Google OAuth2.0 API for CertPathway using the Flask requests library, allowing users to quickly login with a Google account and offloading the responsibility of password management.
- Designed CertPathway's and matching algorithm using the natural language toolkit's wordnet corpus and a precomputed inverted index, leading to matching and loading times of under 150ms during runtime.

NLP Helper Chatbot | *Course Project*

- Created an interactive chatbot that can respond to the user queries with relevant textbook pages using Scikit-Learn's MLPClassifier, a PDF Parser, and wxWidgets.
- Developed a dataset to train our Neural Network Model by randomly recombining sentences from the course's textbook, resulting in a varied dataset that contained 28,800 datapoints, with 576 unique classes.
- Used cross validation to find an optimal set of hyperparameters to achieve a 95% testing accuracy during model runtime.

RecipePlanner | *Personal Project*

- Created an application to make the process of meal-planning and grocery shopping easier by allowing easy selection of recipes that automatically compile into completed lists of the required groceries, including cost & time estimates.
- Developed a GUI using React, allowing a reactive drag-and-drop interface for planning the weeks meal and saving recipes.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Java, SQL, MongoDB, Oracle Database, MATLAB, C#
Frameworks & Libraries: ReactJS, NodeJS, Flask, NumPy, Pandas, Scikit-learn, Matplotlib, wxWidgets, PyQT

EDUCATION

University of Michigan, College of Engineering **December 2023**
Bachelor of Science in Computer Science *Ann Arbor, MI*
Relevant Courses: *Data Structures and Algorithms, Computer Architecture, Human-Centered Design, Cybersecurity, AI, Machine Learning, Web Systems, Database Design, Natural Language Processing, Human-Robot Systems*